Besides, the other two angles namely, $\angle CAB$ and $\angle ACB$ are non-adjacent angles to $\angle CBD$. They are called interior opposite angles of $\angle CBD$.

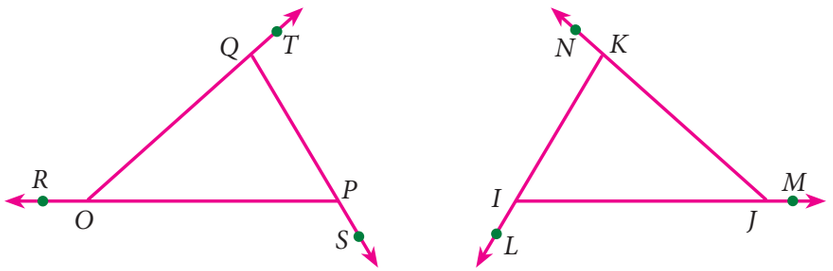
Note

In ΔABC, we can also extend the sides BC to E and CA to F to form exterior angles at C and A.

Exterior Angle Properties of a Triangle

Activity

To understand exeterior angle properties of a triangle list all the exterior angles of triangles that are shown below.



Measure and express each exterior angle as a sum of its interior opposite angles and complete the table.

Exterior angle	Sum of its interior opposite angle

From the above activity we observe that an exterior angle of a triangle is equal to the sum of its interior opposite angles.

Now we prove this result in a formal method.

Proof

In ΔABC, consider, the angles at A, B and C as a, b and c respectively and take exterior angles at A, B and C as x, y and z respectively.

We are going to prove $x=b+c$, $y=a+c$ and $z=a+b$

$a + x = 180°$ [linear pair of angles are supplementary]

This gives, $x = 180° - a$ … (1)

Now, $a + b + c = 180°$ [Sum of 3 angles in a triangle is 180°]

This gives, $b + c = 180° - a$ …(2)

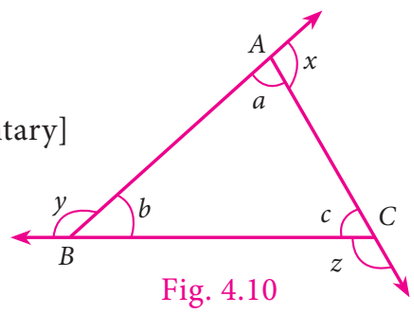


Fig. 4.10

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